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Headphone and operation method thereof

Abstract

Disclosed are a headphone and an operating method thereof. The headphone includes a headband frame, a speaker module, a detector and a controller. The speaker module is disposed on the headband frame. The detector is disposed in the headband frame and detect stress changes on the headband frame to output a frequency response voltage value. The controller is electrically connected to the speaker module and the detector, and receives the frequency response voltage value. The controller determines whether the frequency response voltage value is the same as a target frequency response voltage value to read a target frequency response parameter corresponding to the target frequency response voltage value. The controller drives the speaker module according to the target frequency response parameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan patent application no. 112140812, filed on Oct. 25, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The present disclosure relates to an audio playback device, and in particular to a headphone and an operation method thereof.

Description of Related Art

(3) When using a conventional headphone, due to differences in head sizes of different users, different users might have different hearing experiences, which in turn results in differences in the audio quality received by different users.

SUMMARY

(4) The present disclosure provides a headphone and an operation method thereof, which may provide good audio playback effects.

(5) The headphone of the present disclosure includes a headband frame, a speaker module, a detector and a controller. The speaker module is disposed on the headband frame. The detector is disposed in the headband frame and is disposed to detect the stress change on the headband frame to output a frequency response voltage value. The controller is electrically connected to the speaker module and the detector, and is disposed to receive the frequency response voltage value. The

controller determines whether the frequency response voltage value is the same as the target frequency response voltage value to read the target frequency response parameter corresponding to the target frequency response voltage value to drive the speaker module.

(6) The operation method of the present disclosure is suitable for a headphone. The headphone includes a headband frame, a speaker module, a detector and a controller. The speaker module is disposed on the headband frame, and the detector is disposed in the headband frame. The operation method includes the following steps: detecting the stress change on the headband frame through the detector to output the frequency response voltage value; receiving the frequency response voltage value through the controller; determining whether the frequency response voltage value is the same as the target frequency response voltage value through the controller to read the target frequency response parameter corresponding to the target frequency response voltage value; and driving the speaker module according to the target frequency response parameter through the controller.

(7) Based on the above, the headphone and the operation method thereof in the present disclosure may automatically detect the stress change on the headband frame and adjust the frequency response parameters accordingly to drive the speaker module.

(8) In order to make the above-mentioned features and advantages of the present disclosure more obvious and easy to understand, embodiments are given below and described in detail with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic circuit diagram of a headphone according to an embodiment of the present disclosure.

(2) FIG. 2 is a structural diagram of a headphone according to an embodiment of the present disclosure.

(3) FIG. 3 is a flow chart of an operation method of a headphone according to an embodiment of the present disclosure.

(4) FIG. 4A to FIG. 4D are schematic diagrams of various scenarios of using the headphone according to multiple embodiments of the present disclosure.

(5) FIG. 5 is a schematic diagram of a frequency response curve according to an embodiment of the present disclosure.

(6) FIG. 6 is a flow chart of an operation method of a headphone according to another embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

(7) Reference will now be made in detail to the exemplary embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings. Whenever possible, the same reference numbers are used in the drawings and descriptions to refer to the same or similar parts.

(8) FIG. 1 is a schematic circuit diagram of a headphone according to an embodiment of the present disclosure. Referring to FIG. 1, the headphone **100** includes a controller **110**, a detector **120**, a speaker module **130** and a database **140**. The controller **110** is electrically connected to the detector **120**, the speaker module **130** and the database **140**. In this embodiment, the controller **110** may be a control circuit, such as a microcontroller unit (MCU), a digital signal processor (DSP), a system in a chip (SoC) or the like. The detector **120** may be a stress detector or a force-sensitive resistor (FSR). The speaker module **130** includes a first speaker **131** and a second speaker **132**, and the first speaker **131** and the second speaker **132** are electrically connected to the controller **110**. The database **140** may be implemented as a memory or a related storage device, or may also be implemented as a memory device provided in the controller **110**. In this embodiment, the database **140** may store the target frequency response voltage value, the target frequency response

parameters, the built-in frequency response voltage value and the built-in frequency response parameters described in various embodiments of the present disclosure.

(9) FIG. 2 is a structural diagram of a headphone according to an embodiment of the present disclosure. Referring to FIG. 1 and FIG. 2, in this embodiment, the headphone **100** further includes a headband frame **150**, a first stretching mechanism **161**, a second stretching mechanism **162**, a first earmuff **171** and a second earmuff **172**. The first earmuff **171** is disposed at one end of the headband frame **150** through the first stretching mechanism **161**. The second earmuff **172** is disposed at the other end of the headband frame **150** through the second stretching mechanism **162**. In this embodiment, the detector **120** may be disposed in the headband region **151** of the headband frame **150**, and is disposed to detect the stress change on the headband frame **150**, and output the corresponding voltage value to the controller **110** based on the related circuit matched with the detector **120**, wherein the corresponding voltage value may serve as the frequency response voltage value referred to in various embodiments of the present disclosure. The first speaker **131** may be disposed in the first earmuff **171**. The second speaker **132** may be disposed in the second earmuff **172**. The controller **110** may drive the first speaker **131** and the second speaker **132** to play audio according to the frequency response parameter. In addition, the controller **110** and the database **140** may be disposed in the headband frame **150**, or in the first earmuff **171** or the second earmuff **172**, and the present disclosure is not limited thereto.

(10) FIG. 3 is a flow chart of an operation method of a headphone according to an embodiment of the present disclosure. Referring to FIG. 1 to FIG. 3, in this embodiment, when the user wears the headphone **100**, the headphone **100** may perform operations as described in the following steps **S310** to **S340**. In step **S310**, the detector **120** may detect the stress change on the headband frame **150** to output a frequency response voltage value. The stress change on the headband frame **150** refers to the stress change caused by the deformation of the headband frame **150** when being worn by the user.

(11) In this regard, assuming that the width of the user's head (such as width of cheek or distance between ears) is wider, the distance between the first earmuff **171** and the second earmuff **172** is long, and therefore the stress applied on the headband frame **150** is also high. In contrast, if the width of the user's head is narrow, the distance between the first earmuff **171** and the second earmuff **172** is short, and therefore the stress applied on the headband frame **150** is also low. In addition, when the width of head is fixed, assuming that the length of the user's head (such as the distance from the top of the head to the center of the ears) is long, the user may elongate the first stretching mechanism **161** and the second stretching mechanism **162**, and the stress applied on the headband frame **150** may be reduced. In contrast, when the width of the headband is fixed, assuming that the length of the user's head is short, the user may shorten the first stretching mechanism **161** and the second stretching mechanism **162**, and the stress applied on the headband frame **150** may be increased.

(12) In other words, when different users have different head widths and/or head lengths, the stress caused to the headband frame **150** varies when different users wear the headphone **100**. More importantly, since the extent of deformation of the headband frame **150** is related to how much the first earmuff **171** and the second earmuff **172** fit the user's ears respectively, from another perspective, the stress caused to the headband frame **150** changes along with the extent to which the first earmuff **171** and the second earmuff **172** fit the user's ears respectively. Therefore, the headphone **100** of this embodiment may use the stress change caused to the headband frame **150** as an adjustment basis for determining the frequency response parameters of the audio playback modes of the first speaker **131** and the second speaker **132**.

(13) In step **S320**, the controller **110** may receive a frequency response voltage value corresponding to the stress level of the current headband frame **150** from the detector **120**. In step **S330**, the controller **110** may determine whether the frequency response voltage value is the same as the target frequency response voltage value to read the target frequency response parameter

corresponding to the target frequency response voltage value. If the frequency response voltage value is the same as the target frequency response voltage value, in step **S340**, the controller **110** may drive the first speaker **131** and the second speaker **132** of the speaker module **130** according to the target frequency response parameter.

(14) Specifically, the target frequency response parameter may be an ideal frequency response parameter, which means that the first speaker **131** and the second speaker **132** may be operated in a specific audio playback mode so that the preset user subject is able to receive an ideal audio quality. The controller **110** may compare whether the current frequency response voltage value is the same as the target frequency response voltage value. If they are the same, the controller **110** may directly use the target frequency response parameter corresponding to the target frequency response voltage value to drive the first speaker **131** and the second speaker **132**. If they are not the same, the controller **110** may read the database **140** to select the built-in frequency response parameter corresponding to the built-in frequency response voltage value that is closest to or the same as the frequency response voltage value to drive the first speaker **131** and the second speaker **132** of the speaker module **130** to adjust the audio playback modes of the first speaker **131** and the second speaker **132** accordingly, so that the audio quality received by the user through the first speaker **131** and the second speaker **132** may be close to or equal to the ideal audio quality.

(15) FIG. 4A to FIG. 4D are schematic diagrams of various scenarios of using the headphone according to multiple embodiments of the present disclosure. Referring to FIG. 4A, assuming that the head width of the user wearing the headphone **100** is narrow (the head width is, for example, 132 millimeters (mm)) and the head length of the user is short, when the first stretching mechanism **161** and the second stretching mechanism **162** are in an unstretched state, there is a first height HC1 (for example, 122 mm) from the headband center point P1 to the earmuff center point P2, and there is a distance L1 between the first earmuff **171** and the second earmuff **172**.

(16) Referring to FIG. 4B, assuming that the head width of the user wearing the headphone **100** is wide (for example, the head width is 169 mm) and the head length of the user is short, when the first stretching mechanism **161** and the second stretching mechanism **162** are in an unstretched state, there is a first height HC1 (for example, 122 mm) from the headband center point P1 to the earmuff center point P2, and there is a distance L2 between the first earmuff **171** and the second earmuff **172**.

(17) Referring to FIG. 4C, assuming that the head width of the user wearing the headphone **100** is narrow (for example, the head width is 132 mm) and the head length of the user is long, when the first stretching mechanism **161** and the second stretching mechanism **162** are in a stretched state, there is a second height HC2 (for example, 146 mm) from the headband center point P1 to the earmuff center point P2, and there is a distance L3 between the first earmuff **171** and the second earmuff **172**.

(18) Referring to FIG. 4D, assuming that the head width of the user wearing the headphone **100** is wide (for example, the head width is 169 mm) and the head length of the user is long, when the first stretching mechanism **161** and the second stretching mechanism **162** are in a stretched state, there is a second height HC2 (for example, 146 mm) from the headband center point P1 to the earmuff center point P2, and there is a distance L4 between the first earmuff **171** and the second earmuff **172**.

(19) In light of the foregoing, as shown in FIG. 4A to FIG. 4D, in the above four scenarios, the distance L2 may be a maximum distance, and the distance L3 may be a minimum distance. The distance L2 is greater than the distance L4. The distance L4 is greater than the distance L1. The distance L1 is greater than the distance L3. Therefore, it can be known that the headband frame **150** may have a maximum stress value in the application scenario shown in FIG. 4B (that is, the clamping force generated by the headband frame **150**, the first earmuff **171** and the second earmuff **172** is relatively large). The headband frame **150** may have the second largest stress value in the application scenario shown in FIG. 4D (that is, the clamping force generated by the headband

frame **150**, the first earmuff **171** and the second earmuff **172** is the second largest). The headband frame **150** may have the third largest stress value in the application scenario shown in FIG. **4A** (that is, the clamping force generated by the headband frame **150**, the first earmuff **171** and the second earmuff **172** is the third largest). The headband frame **150** may have the smallest stress value in the application scenario shown in FIG. **4C** (that is, the clamping force generated by the headband frame **150**, the first earmuff **171** and the second earmuff **172** is the smallest). In this way, the controller **110** may determine whether to drive the first speaker **131** and the second speaker **132** according to the target frequency response parameter or the built-in frequency response parameter based on the frequency response voltage value output by the detector **120**.

(20) Further, refer to FIG. **5**, which is a schematic diagram of a frequency response curve according to an embodiment of the present disclosure. For example, the target frequency response parameter may make the audio played by the first speaker **131** and the second speaker **132** heard by the user to have an ideal frequency response curve **501** as shown in FIG. **5**. However, in the case where the clamping force generated by the headband frame **150**, the first earmuff **171** and the second earmuff **172** is relatively large, the frequency response parameter may correspond to the frequency response curve **502** as shown in FIG. **5**. In this regard, since the sound leaks less from the front cavities of the first earmuff **171** and the second earmuff **172**, the user might experience the low frequency more significantly when listening to audio. Therefore, the controller **110** may change the currently used frequency response parameter (adjusted through the built-in frequency response parameter) to moderately reduce the frequency response curve of the low frequency, so that the frequency response curve of the audio heard by the user is able to be close to the ideal frequency response curve **501**.

(21) In contrast, when the clamping force generated by the headband frame **150**, the first earmuff **171** and the second earmuff **172** is relatively small, the frequency response parameter may correspond to the frequency response curve **503** as shown in FIG. **5**. In this regard, since the sound leaks more from the front cavities of the first earmuff **171** and the second earmuff **172**, the user might experience the low frequency less significantly when listening to audio. Therefore, the controller **110** may change the currently used frequency response parameter (adjusted through another built-in frequency response parameter) to moderately increase the frequency response curve of the low frequency, so that the frequency response curve of the audio heard by the user is able to be close to the ideal frequency response curve **501**.

(22) In addition, regarding the position design of the headband region **151** on the headband frame **150**, please refer to FIG. **4A** to FIG. **4D**. When the first stretching mechanism **161** and the second stretching mechanism **162** in FIG. **4A** and FIG. **4B** are in the unstretched state, the deformation amount of the headband frame **150** is concentrated on both sides of the headband center point **P1**, so the headband region **151** where the detector **120** is disposed may be preferably the proportional region corresponding to the first reference height **R1** of the headband frame **150** from the headband center point **P1** to the earmuff center point **P2** between the first earmuff **171** and the second earmuff **172**. The first reference height **R1** is equal to the result of multiplying the first height **HC1** by the preset ratio. Moreover, when the first stretching mechanism **161** and the second stretching mechanism **162** in FIG. **4C** and FIG. **4D** are in the stretched state, the deformation amount of the headband frame **150** is also concentrated on both sides of the headband center point **P1**, so the headband region **151** where the detector **120** is disposed may be preferably the proportional region corresponding to the second reference height **R2** of the headband frame **150** from the headband center point **P1** to the earmuff center point **P2** between the first earmuff **171** and the second earmuff **172**. The second reference height **R2** is equal to the result of multiplying the second height **HC2** by the preset ratio.

(23) Therefore, based on the difference between the unstretched state and the stretched state of the first stretching mechanism **161** and the second stretching mechanism **162**, the headband region **151** where the detector **120** is disposed in this embodiment may be the preset proportional region

corresponding to the preset height of the headband frame **150** from the headband center point **P1** to the earmuff center point **P2**. The preset height may be between (selectable from) the first reference height **R1** and the second reference height **R2**. In an embodiment, the first height **HC1** may be 122 mm, the second height **HC2** may be 146 mm, and the preset ratio is 20%, but the disclosure is not limited thereto.

(24) FIG. **6** is a flow chart of an operation method of a headphone according to another embodiment of the present disclosure. Referring to FIG. **1**, FIG. **2** and FIG. **6**, in step **S610**, the detector **120** may detect the stress change caused to the headband frame **150** to obtain the frequency response voltage value. The detector **120** may output the frequency response voltage value to the controller **110**. In step **S620**, the detector **120** determines whether the frequency response voltage value is greater than or equal to the voltage threshold to determine whether to further determine whether the frequency response voltage value is the same as the target frequency response voltage value. Specifically, the controller **110** may verify whether the user is wearing the headphone **100** by judging whether the frequency response voltage value is greater than or equal to the voltage threshold. It should be noted that the voltage threshold may be, for example, the preset minimum voltage value (i.e., as the trigger voltage value) that is generated by the detector **120** through detection when the headphone **100** is being worn during the product manufacturing process. If the frequency response voltage value is not greater than or equal to the voltage threshold, the controller **110** may re-execute step **S610**. If the frequency response voltage value is greater than or equal to the voltage threshold, in step **S630**, the controller **110** may further determine whether the frequency response voltage value is the same as the target frequency response voltage value.

(25) If the frequency response voltage value is the same as the target frequency response voltage value, in step **S640**, the controller **110** may employ the target frequency response parameter corresponding to the target frequency response voltage value to drive the first speaker **131** and the second speaker **132** of the speaker module **130**. Otherwise, in step **S650**, the controller **110** may select the built-in frequency response parameter corresponding to the built-in frequency response voltage value that is closest to or the same as the frequency response voltage value from the database **140** to drive the speaker module **130** to complete the adaptation adjustment. Therefore, the headphone **100** and the operating method thereof in this embodiment may effectively and automatically adjust the audio quality provided by the first speaker **131** and the second speaker **132** of the speaker module **130**.

(26) It should be noted that the database **140** may be established during the product design and manufacturing process, for example. The database **140** may record multiple built-in frequency response voltage values and multiple built-in frequency response parameters corresponding to the multiple built-in frequency response voltage values. In this regard, under the circumstances where the headband frame **150** applies different clamping forces, the voltage values output by the detector **120** may be collected as multiple built-in frequency response voltage values, and the multiple frequency response parameters of the optimal audio quality played by the first speaker **131** and the second speaker **132** of the speaker module **130** may be collected as the multiple built-in frequency response parameters.

(27) In addition, the target frequency response voltage value may be, for example, the voltage value output by the detector **120** when the headband frame **150** clamps a B&K Type 4128C head and torso simulator (HATS), and the target frequency response parameter may be, for example, obtained by converting the frequency response curve of the optimal audio quality (or specific audio quality) played by the first speaker **131** and the second speaker **132** when the headband frame **150** clamps a B&K Type 4128C HATS, but the disclosure is not limited thereto.

(28) In summary, the headphone and the operation method thereof in the present disclosure may correspondingly and automatically adjust the frequency response parameters for driving the speaker module by detecting the stress changes caused to the headband frame, so that different

users with different head sizes are able to receive similar or better audio playback quality.

(29) Finally, it should be noted that: the above embodiments are only used to illustrate the technical solution of the present disclosure, but not to limit the disclosure; although the present disclosure has been described in detail with reference to the foregoing embodiments, those of ordinary skill in the art should understand that the technical solutions described in the foregoing embodiments can still be modified, or some or all of the technical features can be equivalently replaced; and these modifications or replacement do not deviate the essence of the corresponding technical solutions from the scope of the technical solutions to be protected by the embodiments of the present disclosure.

Claims

1. A headphone comprising: a headband frame; a speaker module disposed on the headband frame; a detector disposed in the headband frame and disposed to detect a stress change on the headband frame to output a frequency response voltage value; and a controller electrically connected to the speaker module and the detector, and disposed to receive the frequency response voltage value, wherein the controller determines whether the frequency response voltage value is the same as a target frequency response voltage value to read a target frequency response parameter corresponding to the target frequency response voltage value to drive the speaker module.
2. The headphone according to claim 1, wherein in response to the controller determining that the frequency response voltage value is the same as the target frequency response voltage value, the controller drives the speaker module using the target frequency response parameter corresponding to the target frequency response voltage value, wherein in response to the controller determining that the frequency response voltage value is not the same as the target frequency response voltage value, the controller selects a built-in frequency response parameter of a built-in frequency response voltage value that is closest to or the same as the frequency response voltage value from a database to drive the speaker module.
3. The headphone according to claim 1, wherein the controller pre-determines whether the frequency response voltage value is greater than or equal to a voltage threshold to determine whether to further determine whether the frequency response voltage value is the same as the target frequency response voltage value.
4. The headphone according to claim 1, further comprising: a first stretching mechanism; a first earmuff disposed at one end of the headband frame through the first stretching mechanism; a second stretching mechanism; and a second earmuff disposed at the other end of the headband frame through the second stretching mechanism, wherein the detector is disposed in a headband region of the headband frame, and the headband region is a preset proportional region corresponding to a preset height of the headband frame from a headband center point to an earmuff center point between the first earmuff and the second earmuff.
5. The headphone according to claim 4, wherein when the first stretching mechanism and the second stretching mechanism are in an unstretched state, there is a first height from the headband center point to the earmuff center point, and when the first stretching mechanism and the second stretching mechanism are in a completely stretched state, there is a second height from the headband center point to the earmuff center point, wherein the preset height is between a first reference height and a second reference height, the first reference height is equal to a result of multiplying the first height by a preset ratio, and the second reference height is equal to a result of multiplying the second height by the preset ratio.
6. The headphone according to claim 5, wherein the first height is 122 mm, and the second height is 146 mm.
7. The headphone according to claim 5, wherein the preset ratio is 20%.
8. The headphone according to claim 4, wherein the speaker module comprises: a first speaker

electrically connected to the controller, and disposed in the first earmuff; and a second speaker electrically connected to the controller, and disposed in the second earmuff.

9. The headphone according to claim 1, wherein the detector is a stress detector.

10. The headphone according to claim 1, wherein the detector is a force-sensitive resistor.

11. An operation method of a headphone, wherein the headphone comprises a headband frame, a speaker module, a detector and a controller, wherein the speaker module is disposed on the headband frame, and the detector is disposed in the headband frame, wherein the operation method comprises: detecting a stress change on the headband frame through the detector to output a frequency response voltage value; receiving the frequency response voltage value through the controller; determining whether the frequency response voltage value is the same as a target frequency response voltage value through the controller to read a target frequency response parameter corresponding to the target frequency response voltage value; and driving the speaker module according to the target frequency response parameter through the controller.

12. The operation method according to claim 11, wherein determining whether the frequency response voltage value is the same as the target frequency response voltage value through the controller comprises: in response to the controller determining that the frequency response voltage value is the same as the target frequency response voltage value, the controller drives the speaker module using the target frequency response parameter corresponding to the target frequency response voltage value; and in response to the controller determining that the frequency response voltage value is not the same as the target frequency response voltage value, the controller selects a built-in frequency response parameter of a built-in frequency response voltage value that is closest to or the same as the frequency response voltage value from a database to drive the speaker module.

13. The operation method according to claim 11, further comprising: pre-determining whether the frequency response voltage value is greater than or equal to a voltage threshold through the controller to determine whether to further determine whether the frequency response voltage value is the same as the target frequency response voltage value.

14. The operation method according to claim 11, wherein the headphone further comprises a first earmuff, a second earmuff, a first stretching mechanism and a second stretching mechanism, wherein the first earmuff is disposed at one end of the headband frame through the first stretching mechanism, and the second earmuff is disposed at the other end of the headband frame through the second stretching mechanism, wherein the detector is disposed in a headband region of the headband frame, and the headband region is a preset proportional region corresponding to a preset height of the headband frame from a headband center point to an earmuff center point between the first earmuff and the second earmuff.

15. The operation method according to claim 14, wherein when the first stretching mechanism and the second stretching mechanism are in an unstretched state, there is a first height from the headband center point to the earmuff center point, and when the first stretching mechanism and the second stretching mechanism are in a completely stretched state, there is a second height from the headband center point to the earmuff center point, wherein the preset height is between a first reference height and a second reference height, the first reference height is equal to a result of multiplying the first height by a preset ratio, and the second reference height is equal to a result of multiplying the second height by the preset ratio.

16. The operation method according to claim 15, wherein the first height is 122 mm, and the second height is 146 mm.

17. The operation method according to claim 15, wherein the preset ratio is 20%.

18. The operation method according to claim 14, wherein the speaker module comprises a first speaker and a second speaker, the first speaker is electrically connected to the controller, and is disposed in the first earmuff, the second speaker is electrically connected to the controller, and is disposed in the second earmuff.

19. The operation method according to claim 11, wherein the detector is a stress detector.
 20. The operation method according to claim 11, wherein the detector is a force-sensitive resistor.
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